Curriculum Vitae

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Universidad Politécnica de Madrid
Escuela Técnica Superior de Ingenieros
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Education

- PhD Mathematics, Central European University (CEU), Budapest, Hungary, 2010-2015.
- MSc Mathematics, Eötvös Loránd Tudományegyetem (ELTE), Budapest, Hungary, 2008-2010.
- BS Mathematics, University of Iceland, Reykjavík, Iceland, 2005-2008.
- Gymnasium, Science line, Menntaskólinn Hraðbraut, Reykjavík, Iceland, 2003-2005.

Thesis

- *The geometric genus and Seiberg–Witten invariant of Newton nondegenerate surface singularities*, Central European University, Advisor: Némethi András, defended December 1, 2015, with summa cum laude, arXiv:1601.02572.

Employment

- 2024-: Profesor Ayudante Doctor at Universidad Politécnica de Madrid.
- 2022-2024: María Zambrano postdoc at Universidad Complutense de Madrid.
- 2021-2022: Simons Postdoc at the Institute of Mathematics, Vietnam Academy of Science and Technology, Hanoi.
- 2018-2020: Postdoc at Universidad Nacional Autónoma de México, Cuernavaca.
- 2018-2018: Juan de la Cierva Postdoc Fellow at Basque Center for Applied Mathematics.
- 2016-2018: ERC Postdoc Fellow at Basque Center for Applied Mathematics.
- 2013-2016: ERC postdoctoral fellow at the Alfréd Rényi Institute for Mathematics in Budapest.

Grants

- Co-PI, jointly with Pablo Portilla Cuadrado, of «Proyecto de generación de conocimiento», financed by the spanish Ministry of Science, Innovation and Universities, reference number PID2025-171502NA-I00.

Preprints

- Joint with Juan Viu-Sos, The Milnor fiber boundary of an arrangement determines its combinatorics, arXiv:2512.05891.
- Joint with Pablo Portilla Cuadrado, On the integral variation map of isolated plane curves singularities, arXiv:2512.05820.

Accepted for publication

- Joint with Pablo Portilla Cuadrado, The total spine of the Milnor fibration of a plane curve singularity, accepted for publication in the Memoirs of the American Mathematical Society, 2025, arXiv:2305.12555.
- Joint with András Némethi, Local Newton nondegenerate Weil divisors in toric varieties, accepted for publication in the Michigan Mathematical Journal, 2024, arXiv:2102.02948.

Published articles

- Jacobian polygon and Łojasiewicz exponent of isolated complex hypersurface singularities, Advances in Mathematics, Vol. 495, doi:10.1016/j.aim.2026.110955.
- On ideal filtrations for Newton nondegenerate surface singularities, Singularities and their interaction with geometry and low dimensional topology. In honor of András Némethi on the occasion of his 60th birthday. Selected papers based on the presentations at the conference "Némethi60: geometry and topology of singularities", Budapest, Hungary, May 2731, 2019, Basel: Birkhäuser/Springer. Trends Math., 13-32 (2021). doi:10.1007/978-3-030-61958-9_2.
- Joint with Pablo Portilla Cuadrado, Mixed tête-à-tête twists as monodromies associated with holomorphic function germs, Geom. Dedicata 210, 43-64 (2021), doi:10.1007/s10711-020-00533-7.
- The boundary of the Milnor fiber of the singularity $f(x,y) + zg(x,y) = 0$, Eur. J. Math. 6, No. 4, 1191-1215 (2020), doi:10.1007/s40879-019-00393-w.
- Joint with András Némethi, The geometric genus of hypersurface singularities, J. Eur. Math. Soc. (JEMS) 18, No. 4, 825-851 (2016), doi:10.4171/JEMS/604.
- The Milnor fiber of the singularity $f(x,y) + zg(x,y) = 0$. Rev. Mat. Complut. (2016) 29: 225, doi:10.1007/s13163-015-0179-5.

Conference talks

- On the Jacobian polygon and Łojasiewicz exponent of isolated complex hypersurface singularities, Łojasiewicz100, Jagiellonian University, Kraków, Poland, September 21-25, 2026.
- The total spine of the Milnor fibration of a plane curve singularity, Singularities, Oberwolfach Research Institute Oberwolfach, Germany, September 28 - October 3, 2025.
- The BKO conjecture on the Łojasiewicz exponent for Newton nondegenerate hypersurface singularities, 9th annual Meeting of the Spanish Network for Algebraic Geometry and Singularity Theory Universidad Politécnica de Madrid, January 13-15, 2025.
- The total spine and invariant Milnor fibration at radius zero for plane curve singularities, Congreso Bienal de la Real Sociedad Matemática Española Real Sociedad Matemática Española Pamplona, Spain, January 22-26, 2024.
- The total spine of the Milnor fibration of a plane curve, The Interactive Workshop, part of the research semester "Singularities and low dimensional topology", Erdős center, Budapest, Hungary, June 19-23, 2023.
- Tropical bases and Cohen-Macaulayification, Workshop on Algebraic Geometry and Singularity Theory, Vietnam Institute for Advanced Study in Matematics, Ha Long, Vietnam, April 24, 2022.
- Local Newton nondegenerate Weil divisors in toric varieties, Singularities, Oberwolfach Research Institute Oberwolfach, Germany, September 26 - October 2, 2021.
- The geometric genus and Seiberg–Witten invariant of Newton nondegenerate Weil divisors in a toric space, Workshop on Topological and Analytical Methods in Singularity Theory, CIMAT, Guanajuato, Mexico, November 14, 2019.
- The boundary of the Milnor fiber of the singularity $f(x,y) + zg(x,y) = 0$, Lipschitz Geometry of Singularities, October 23, 2018, Oaxaca, Mexico.
- The geometric genus and Seiberg–Witten invariant of Newton nondegenerate surface singularities, 4ºCongreso de Jóvenes Investigadores, Valencia, Spain, September 4-8, 2017.
- The geometric genus and Seiberg–Witten invariant of Newton nondegenerate surface singularities, Singularities, Oberwolfach Research Institute Oberwolfach, Germany, September 25-October 1, 2016.
- The geometric genus of hypersurfaces, 13th International Workshop on Real and Complex Singularities Celebrating the 60th birthday of María del Carmen Romero Fuster, São Carlos, Brazil, July 27-August 1, 2014.

Seminar presentations

- On the Jacobian polygon and Łojasiewicz exponent of isolated complex hypersurface singularities, Gdańsk-Kraków-Łódź-Warszawa Seminar in Singularity Theory, Online, Poland, January 9, 2026.
- On the Jacobian polygon and Łojasiewicz exponent of isolated complex hypersurface singularities, Seminario de Álgebra, Universidad Autónoma de Madrid, Madrid, Spain, November 17, 2025.
- The total spine of the Milnor fibration of a plane curve, Geometry Seminar, Instituto de Ciencias Matemáticas, Madrid, Spain, October 11, 2023
- Espinas Lagrangianas en fibras de Milnor de singularidades de curvas planas, Seminario de Álgebra, Geometría y Topología, Universidad Complutense de Madrid, Madrid, Spain, May 9, 2023.
- Mixed tête-à-tête twists as monodromies associated with holomorphic function germs, JV Singularity Seminar, JSPS-VAST Bilateral Joint Research Program, Hanoi, Vietnam, March 3, 2022.
- Newton nondegenerate Weil divisors in toric varieties, Séminaire Marseillais de Théorie des Singularités Aix-Marseille University Marseille, France June 17, 2021.
- Newton nondegenerate Weil divisors in toric varieties Iberoamerican Webminar of Young Researchers in Singularity Theory and related topics, Instituto de Matemática Interdisciplinar, January 20, 2021.
- Local tropical Cartier divisors and the multiplicity Tropical Geometry in Frankfurt/Zoom TGiF/Z, Goethe Universität, Frankfurt, Germany (remotely), May 28, 2020.
- Tropical intersection and singularity theory, Seminar on Geometry and Topology, Institute of Mathematics, Vietnam Academy of Science and Technology, Hanoi, Vietnam, May 21, 2020.
- Divisores Weil Newton nodegenerados en variedades tóricas, Seminario de Álgebra Conmutativa y Geometría Algebraica, CIMAT, Guanajuato, Mexico, May 6, 2019.
- The boundary of the Milnor fiber of the hypersurface singularity $f(x, y) + zg(x, y) = 0$, Séminaire Singularités, FRUMAM, Université Aix-Marseille, Marseille, France, November 23, 2017.
- The boundary of the Milnor fiber of the hypersurface singularity $f(x, y)+zg(x, y) = 0$, Seminar, Géométrie des espaces singuliers, Laboratoire Paul Painlevé, Lille, France, December 13, 2016.
- The geometric genus and Seiberg–Witten invariant of Newton nondegenerate surface singularities, Algebraic Geometry and Differential Topology Seminar, Alfréd Rényi Institute of Mathematics, Budapest, Hungary, October 16, 2015.
- The geometric genus of hypersurface singularities, Algebraic Geometry and Differential Topology Seminar, Alfréd Rényi Institute of Mathematics, Budapest, Hungary, March 14, 2014.

Visits

- Budapest, Hungary, March 1 - April 30, 2023, participant in semester on Singularities and Low Dimensional Topology, organized by the Erdős center at the Alfréd Rényi Institute of Mathematics.
- Hanoi, Vietnam, March 1-29, 2020, invited by Lê Quý Thường at VIASM.
- Marseille, France, November 21-25, 2017, invited by Anne Pichon at Institut de Mathématiques de Marseille.
- Lille, France, December 11-15, 2016, invited by Patrick Popescu-Pampu at Le laboratoire de mathématiques Paul Painlevé, Université de Lille.
- Budapest, Hungary, April 17-23, 2016, invited by András Némethi at the Alfréd Rényi Institute of Mathematics.

Learning seminar talks

- Introduction to Heegaard-Floer homology, Seminario de Geometría Algebraica y Singularidades, BCAM, May 5 and 26, 2017, Bilbao, Spain.
- Newton nondegenerate surface singularities, Seminario de Geometría Algebraica y Singularidades, BCAM, November 11, December 1, 2016, Bilbao, Spain.

- Introduction to normal surface singularities, Seminario de Geometría Algebraica y Singularidades, BCAM, October 21, 2016, Bilbao, Spain.
- The spectrum of hypersurface singularities, Seminario de Geometría Algebraica y Singularidades, BCAM, May 2016, Bilbao, Spain.
- Mixed Hodge structures on cohomologies of smoooth varieties, Singularity theory and low dimensional topology seminar, Alfréd Rényi Institute of Mathematics, March 19 and 26, 2013, Budapest, Hungary.
- Basic Hodge theory, Singularity theory and low dimensional topology seminar, Alfréd Rényi Institute of Mathematics, November 29 and December 6, 2012, Budapest, Hungary.

Conferences attended

- Łojasiewicz100, Jagiellonian University, Kraków, Poland, September 21-25, 2026.
- Singularities, Oberwolfach Research Institute Oberwolfach, Germany, September 28 - October 3, 2025.
- Singularities in Plovdiv, ICMS-Sofia, Plovdiv, Bulgaria, June 8-13, 2025.
- 9th annual Meeting of the Spanish Network for Algebraic Geometry and Singularity Theory Universidad Politécnica de Madrid, January 13-15, 2025.
- Congreso Bienal de la Real Sociedad Matemática Española Real Sociedad Matemática Española Pamplona, Spain, January 22-26, 2024.
- The Interactive Workshop, part of the research semester "Singularities and low dimensional topology", Erdős center, Budapest, Hungary, June 19-23, 2023.
- Algebraic and topological interplay of algebraic varieties A Conference in Honor of the 60th Birthday of Enrique Artal and the 55th Birthday of Alejandro Melle, Jaca, Spain, June 12-16, 2023.
- The Low-dimensional Workshop, part of the research semester "Singularities and low dimensional topology", Erdős center, Budapest, Hungary, March 27-31, 2023.
- Workshop on Algebraic Geometry and Singularity Theory, Vietnam Institute for Advanced Study in Matematics, Ha Long, Vietnam, April 24, 2022.
- Singularities, Oberwolfach Research Institute Oberwolfach, Germany, September 26 - October 2, 2021.
- Workshop on Topological and Analytical Methods in Singularity Theory, CIMAT, Guanajuato, Mexico, November 4-15, 2019.
- BCM-Singularies 4th meeting on singularities Brazil-Chile-Mexico, Universidad de Talca, Talca, Chile, March 11-15, 2019.
- Tropical Methods in Real Algebraic Geometry Casa Matemática Oaxaca, Oaxaca, Mexico, September 8-13, 2019.
- Non-Isolated Singularities and derived Geometry, A celebration of the 60th birthday of David Massey, UNAM, Cuernavaca, Mexico, July 29 - August 3, 2019.
- 4th meeting on singularities Brazil-Chile-Mexico, Universidad de Talca, Talca, Chile March 11-15, 2019.
- Lipschitz Geometry of Singularities, Casa Matemática Oaxaca, Oaxaca, Mexico. October 21-26, 2018.
- 15th International Workshop on Real and Complex Singularities, Universidade de São Paulo, Campus São Carlos, São Carlos, Brazil, July 22-28, 2018,
- 4ºCongreso de Jóvenes Investigadores, Real Sociedad Matemática Española, Universitat de València, Valencia, Spain, September 4-8, 2017.
- A Panorama on Singular Varieties: A conference to celebrate Lê Dũng Tráng's 70th Birthday, IMUS, Sevilla, Spain, February 7-10, 2017.
- Singularities, Oberwolfach, Germany, September 26-30, 2016.
- Conference of the Belgian, Royal Spanish and Luxembourg Mathematical Societies, Logroño, Spain, June 6-8, 2016.

- Algebra, Geometry and Topology of Singularities, Galatasaray University, Istanbul, Turkey May 8-12, 2016.
- Local and global Invariants of Singularities, En l'honneur de Jean-Paul Brasselet à l'occasion de son 70ème anniversaire, CIRM, Luminy, France, February 23-27, 2015.
- Workshop on Singularities in geometry, topology, foliations and dynamics 2014, A celebration of the 60th birthday of Pepe Seade, Universidad Autónoma de Yucatan, Mérida, Mexico, December 8-19, 2014.
- 13th International Workshop on Real and Complex Singularities, Celebrating the 60th birthday of María del Carmen Romero Fuster, ICMC, São Carlos, Brazil, July 27-August 1, 2014.
- Algebra and Geometry and Topology of Singularities, On the occasion of the 60th birthday of I. Luengo, La Cristalera, Miraflores de la Sierra, Spain, September 9-13, 2013.
- Singularities in geometry and applications III, ICMS, Edinburgh, UK, September 02-06, 2013.
- Geometry and topology of smooth 4-manifolds, Max Planck Institute for Mathematics, Bonn, Germany, June 3-7, 2013.
- Geometry and topology of complex singularities, CIRM, Luminy, France, April 15-19, 2013.
- Conference on singularities and geometry-topology in honour of Sabir M. Gusein-Zade with occasion of his 60th birthday, El Escorial, Madrid, Spain, October 10-16, 2010.

Schools attended

- Introduction To Geometric Analysis: The Atiyah-Singer Index Theorem, BCAM-UPV/EHU, Bilbao, Spain, June 5-23, 2017.
- Deformations in Turin, University of Turin, Turin, Italy, July 5-8, 2017.
- Workshop on Algebra, Geometry and Topology of Singularities, Gebze Technical University, Istanbul, Turkey, May 2-7, 2016.
- School on Singularity theory in topology, geometry and foliations, Instituto de Matemáticas (Unidad Cuernavaca) of the Universidad Nacional Autónoma de México (UNAM), November 24-December 5, 2014.
- Moduli and Automorphic forms, IRTG and Alfréd Rényi Institute of Mathematics, Budapest, Hungary, August 25-30, 2014.
- Applications of étale cohomology, Alfréd Rényi Institute of Mathematics, Budapest, Hungary, June 16-20, 2014.
- New aspects on Singularity Theory, ICMAT, Madrid, Spain, September 16-27, 2013.
- Topology of singularities, University of Mainz, Mainz, Germany, September 19-23, 2011.
- Singularities: algebra, geometry and applications, Institute of Mathematics of the National Academy of Sciences of Ukraine and Kiev National Taras Shevchenko University, Kiev, Ukraine, August 8-19, 2011.
- School on D-modules and applications in Singularity Theory, CSIC, ICMAT and IMUS, Sevilla, June 20-25 and Madrid June 27-July 2, Spain, 2011.

Teaching Experience

- Álgebra lineal y Geometría Analítica, Universidad Politécnica de Madrid, autumn 2024, 2025 and 2026.
- Cálculo II, Universidad Politécnica de Madrid, spring 2026 (scheduled for 2027 as well).
- Estadística y Optimización, Universidad Politécnica de Madrid, spring 2025.
- Curvas Algebraicas, Universidad Complutense de Madrid, autumn 2022 and 2023.

Misc

- Award for Advanced Doctoral Students, CEU, Budapest, Hungary, academic year 2012/2013.
- Second honorable mention in the Schweitzer Miklós Memorial Competition 2008, Budapest, Hungary.
- Participant in International Mathematical Olympiad, 3 points, July 8-19, 2005, Mérida, Mexico.
- On Honorary List of students academic year 2004/2005, Hraðbraut Gymnasium, Reykjavík, Iceland.

Software

- `monodromy_var` Jointly with Pablo Portilla Cuadrado. An implementation of our results on the variation map of plane curves. Written in python.
- `jacobian_polygon` Calculates the Jacobian and alternating Jacobian polygons associated with Newton nondegenerate hypersurfaces. Written in Sage.
- `plumbing_graph` An implementation of Neumann's plumbing calculus. Written in Sage.
- `lattice/ latco` An interface and library which calculates the root and Euler characteristics of lattice cohomology defined by Némethi. Written in C,
- `Oka` A collection of functions related to surface singularities, among them an implementation of Oka's algorithm for the resolution of Newton nondegenerate hypersurfaces. Written in PARI/GP.

Languages

- Icelandic, native,
- English, fluent,
- Spanish, C1,
- Danish, Swedish, Hungarian, less than fluent.